



MOIZ DAWOODI
O/A LEVEL PHYSICS

O' LEVEL PHYSICS

Electrical Quantities

4.2

STUDENT NAME



0315-0805775

PHYSICS WITH
MOIZ DAWOODI

SYLLABUS OUTLINE:

4.2 Electrical quantities

4.2.1 Electrical charge

- 1 State that there are positive and negative charges and that charge is measured in coulombs
- 2 State that unlike charges attract and like charges repel
- 3 Describe experiments to show electrostatic charging by friction
- 4 Explain that charging of solids by friction involves only a transfer of negative charge (electrons)
- 5 Describe an electric field as a region in which an electric charge experiences a force
- 6 State that the direction of an electric field line at a point is the direction of the force on a positive charge at that point
- 7 Describe simple electric field patterns, including the direction of the field:
 - (a) around a point charge
 - (b) around a charged conducting sphere
 - (c) between two oppositely charged parallel conducting plates (end effects will **not** be examined)
- 8 State examples of electrical conductors and insulators
- 9 Describe an experiment to distinguish between electrical conductors and insulators
- 10 Recall and use a simple electron model to explain the difference between electrical conductors and insulators

4.2.2 Electrical current

- 1 Define electric current as the charge passing a point per unit time; recall and use the equation

$$\text{electric current} = \frac{\text{charge}}{\text{time}}$$

$$I = \frac{Q}{t}$$
- 2 Describe electrical conduction in metals in terms of the movement of free electrons
- 3 Know that current is measured in amps (amperes) and that the amp is given by coulomb per second (C/s)
- 4 Know the difference between direct current (d.c.) and alternating current (a.c.)
- 5 State that conventional current is from positive to negative and that the flow of free electrons is from negative to positive
- 6 Describe the use of ammeters (analogue and digital) with different ranges

4.2.3 Electromotive force and potential difference

- 1 Define e.m.f. (electromotive force) as the electrical work done by a source in moving a unit charge around a complete circuit; recall and use the equation

$$\text{e.m.f.} = \frac{\text{work done (by a source)}}{\text{charge}}$$

$$E = \frac{W}{Q}$$

continued

4.2 Electrical quantities continued

4.2.3 Electromotive force and potential difference continued

- 2 Define p.d. (potential difference) as the work done by a unit charge passing through a component; recall and use the equation

$$\text{p.d.} = \frac{\text{work done (on a component)}}{\text{charge}}$$

$$V = \frac{W}{Q}$$

- 3 Know that e.m.f. and p.d. are measured in volts and that the volt is given by joule per coulomb (J/C)
- 4 Describe the use of voltmeters (analogue and digital) with different ranges
- 5 Calculate the total e.m.f. where several sources are arranged in series
- 6 State that the e.m.f. of identical sources connected in parallel is equal to the e.m.f. of one of the sources

4.2.4 Resistance

- 1 Recall and use the equation

$$\text{resistance} = \frac{\text{p.d.}}{\text{current}}$$

$$R = \frac{V}{I}$$

- 2 Describe an experiment to determine resistance using a voltmeter and an ammeter and do the appropriate calculations
- 3 Recall and use, for a wire, the direct proportionality between resistance and length, and the inverse proportionality between resistance and cross-sectional area
- 4 State Ohm's law, including reference to constant temperature
- 5 Sketch and explain the current–voltage graphs for a resistor of constant resistance, a filament lamp and a diode
- 6 Describe the effect of temperature increase on the resistance of a resistor, such as the filament in a filament lamp

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CHARGE:

"Charge is the gain of an electron or loss of an electron."

FORMULA:

$$\text{Current} = \frac{\text{Charge}}{\text{Time}}, \quad I = \frac{q}{t}$$

The S.I unit of charge is Coulomb (C).

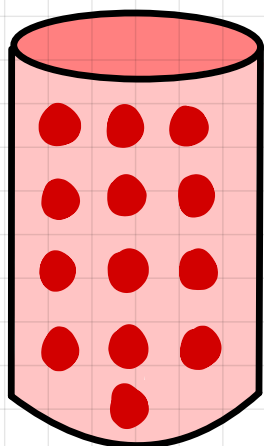
$$q = I t$$

Neutral body has same no. of positive & negative charge.

Positive charge is the absence of an electron.

Coulomb = Ampere x second

CHARGE ANALOGOUS.



Mass of the container is negligible

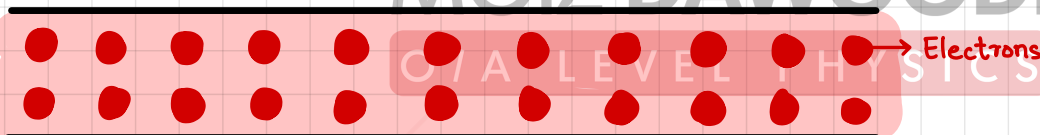
Mass of each ball = $m = 10\text{g}$

Total mass of the ball = $M = 1000\text{g}$

How many balls are there?

$$\text{No. of balls} = \frac{M}{m} = \frac{1000}{10} = 100 \text{ balls}$$

Now,



Charge of single electron = $e = 1.6 \times 10^{-19}\text{C}$

Total charge in wire = $Q = 1\text{C}$

How many electrons are there in wire?

$$\text{no. of electrons} = \frac{Q}{e} = \frac{1}{1.6 \times 10^{-19}} = 6.25 \times 10^{18} \text{ electrons.}$$

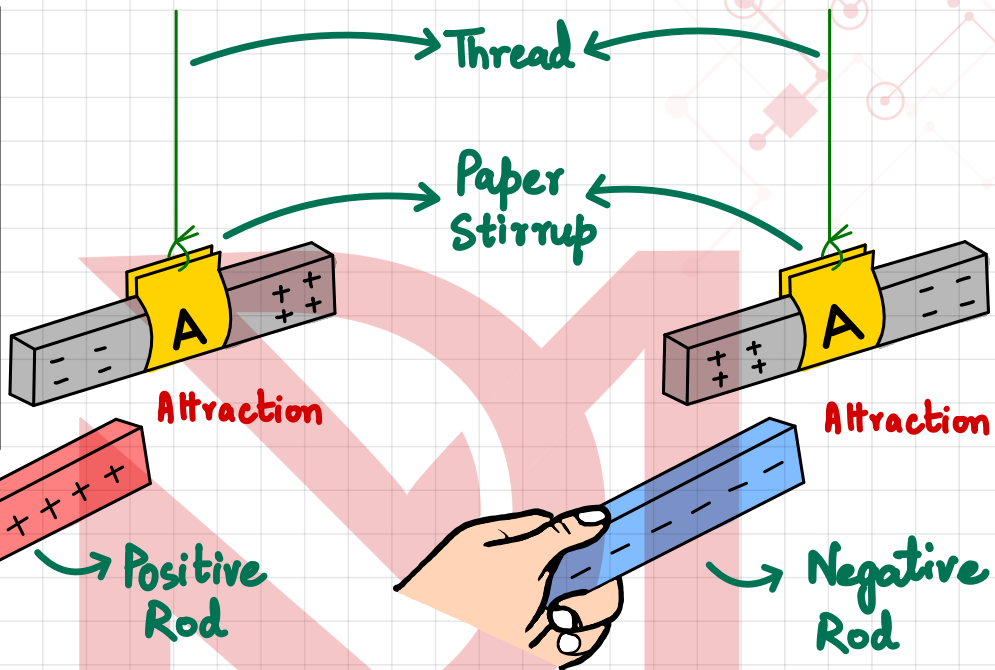
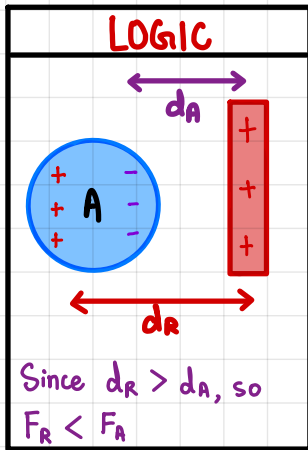
GENERAL FORMULA.

$$n = \frac{Q}{1.6 \times 10^{-19}}$$

LAWS OF ELECTROSTATICS:

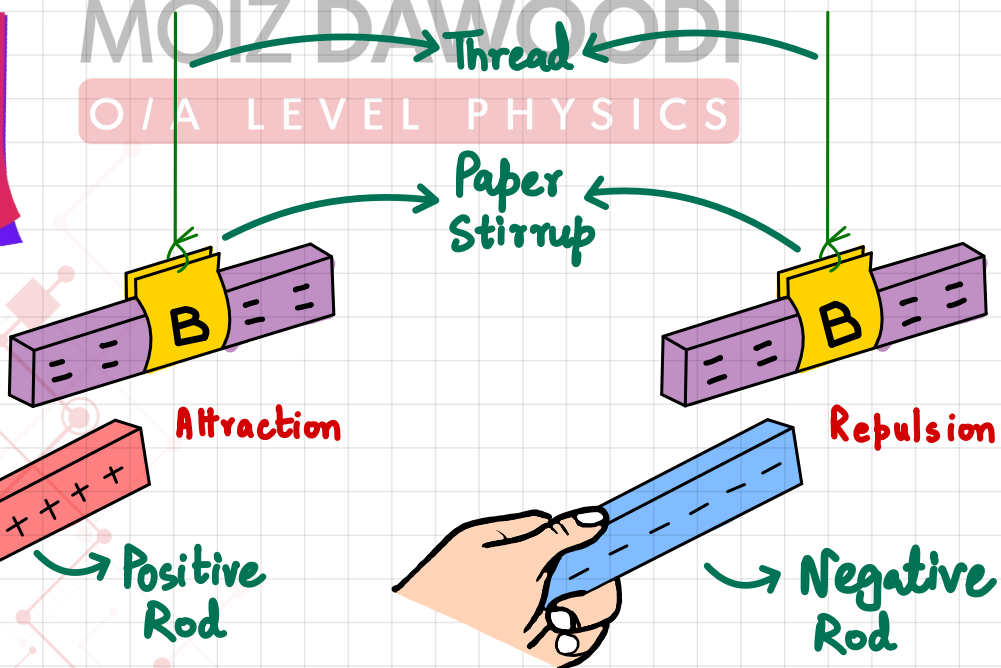
- Like charges repel.
- Unlike charges attract.

IDENTIFICATION OF CHARGE:



Rod A is neutral as it is attracted by both rods. Attraction is greater than repulsion as distance b/w unlike charges is less.

Attraction doesn't help to identify charge, only repulsion helps to identify charge.



Rod B is negatively charged as it is repelled by negative rod (Like charges repel).

CHARGING DUE TO FRICTION:

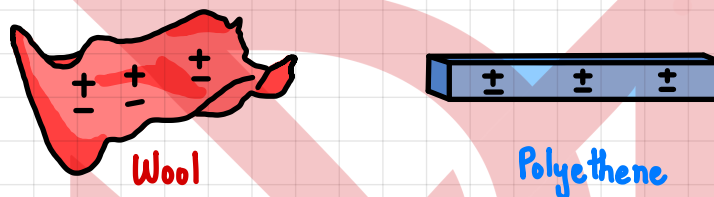
	POSITIVE	NEGATIVE (strong Nucleus)
★ ★	Wool	Polyethene
	Glass	Wool
	Hair	Comb

Charging due to friction is only efficient with insulators.

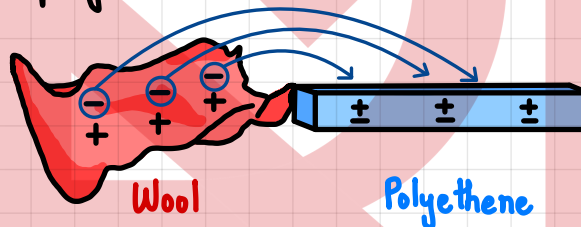
Electron always prefer to move to an atom which has strong nucleus.

Describe how polyethene & wool are charged by friction.

- Bring neutral wool & polyethene.



- Start rubbing wool & polyethene. Electrons of wool receive heat.



- As wool has lost electrons, it becomes positive & polyethene has gained electrons, it becomes negative.



Q: Explain in terms of movement of electrons, how polyethene becomes negative & wool becomes positive charged? [3]

Due to friction, electrons of wool receive heat & jumped towards polyethene. As wool has lost electrons, it becomes positive charged while polyethene has gained electrons, it becomes negative charged.



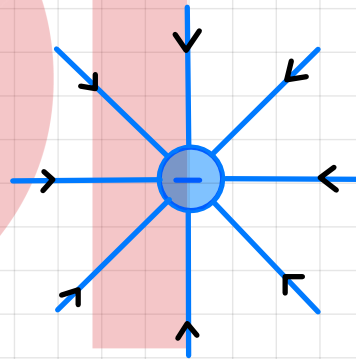
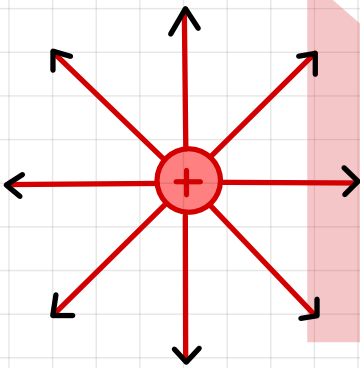
ELECTRIC FIELD.

"Region around a charge in which charge experiences electric force of other charge."

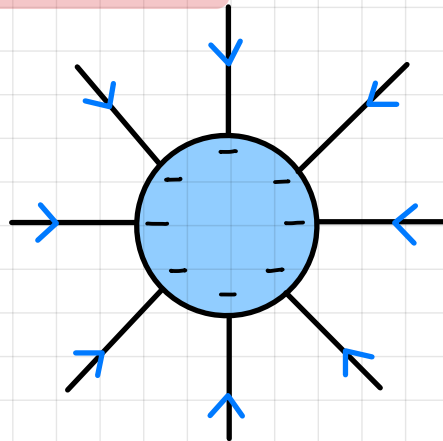
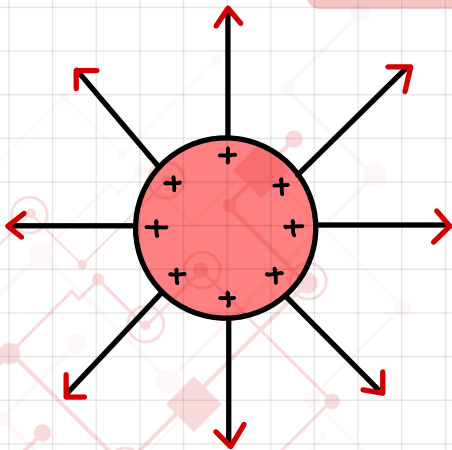
- Direction of electric field for positive charge is outward. $\oplus \rightarrow$
- Direction of electric field for negative charge is inward. $\ominus \leftarrow$
- The direction of electric field is the direction of force on positive charge.

PATTERNS OF ELECTRIC FIELD:

- Around point charge.

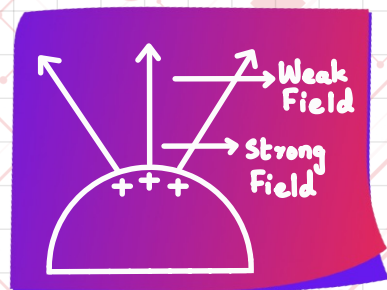
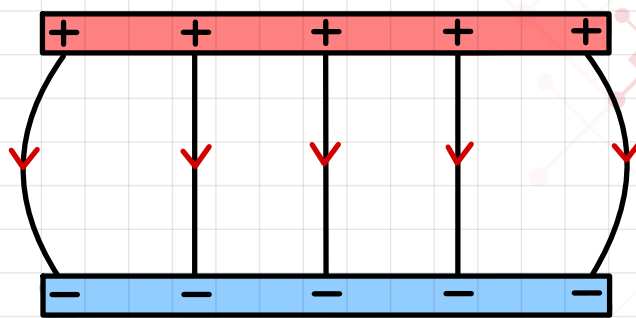


- Around a charged sphere.

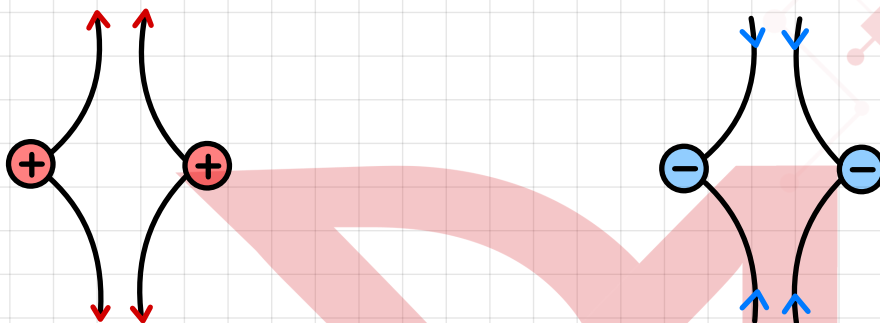


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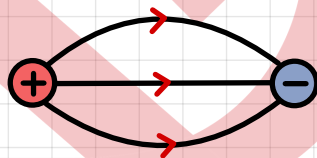
- Between two oppositely charged parallel conducting plates.



- Between like charges



- Between unlike charges



CONDUCTORS & INSULATORS.

When some material gains charge, they lose it immediately This is because electron flow through them until the balance of positive & negative charge is restored.

CONDUCTORS:

"Materials which let current pass through them are called conductors." They have free electrons. Mostly metals are good conductors.

Examples: Iron, copper, Silver etc.

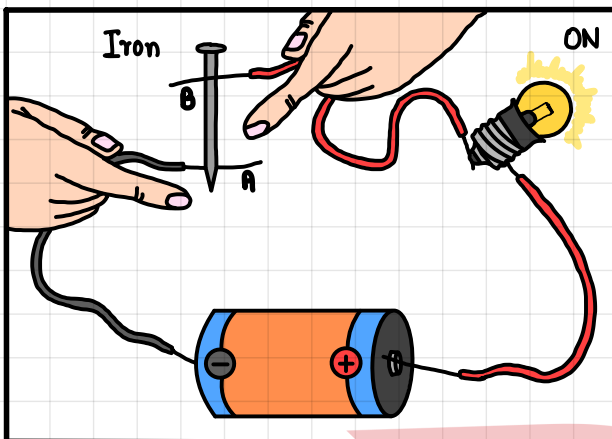
INSULATORS:

"Materials which don't let current pass through them are called insulators."

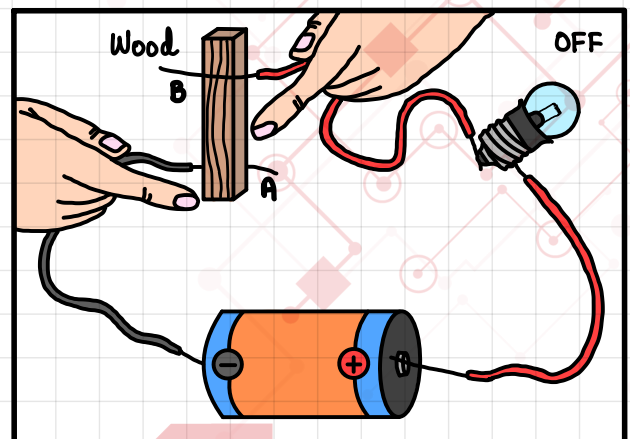
They don't have free electrons. Mostly non-metals are good insulators.

Examples: Plastic, wood, glass etc.

EXPERIMENT TO DISTINGUISH BETWEEN ELECTRICAL CONDUCTOR & INSULATOR:

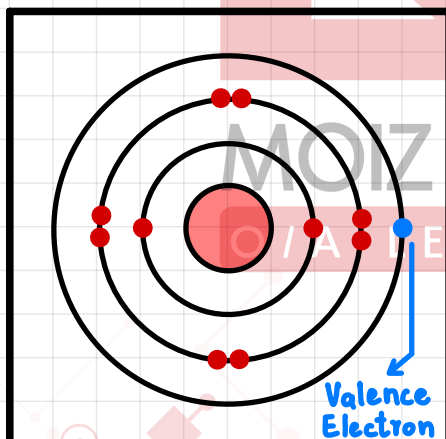


Place iron nail between A & B.
Bulb glows which proves that
iron is a conductor.

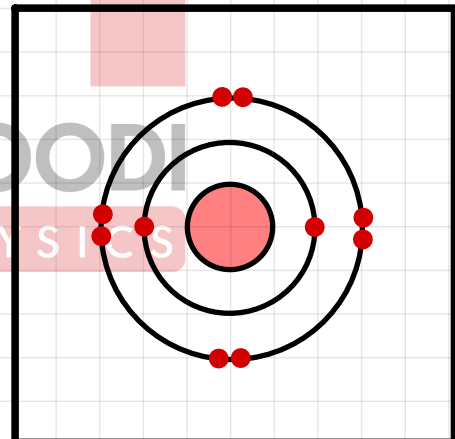


Place wooden bar between A & B.
Bulb doesn't glow which proves
that wood is an insulator.

DIFFERENCE BETWEEN CONDUCTOR & INSULATOR BY ELECTRON MODEL.



Conductors contain
valence electrons
in outermost orbit
& that orbit is
incomplete.



Insulators doesn't
have valence electrons
in outer most orbit
& that orbit is
complete.

CURRENT:

"Current is the flow of charges per unit time."

The S.I unit of current is Ampere (A).

$$\text{Ampere} = \frac{\text{Coulomb}}{\text{Second}}$$

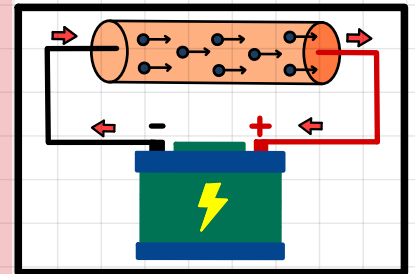
FORMULA:

$$\text{Current} = \frac{\text{Charge}}{\text{Time}}$$

$$I = \frac{q}{t}$$

ELECTRICAL CONDUCTION IN METALS DUE TO FREE ELECTRONS:

Metals contain free electrons. When a piece of wire is connected with a battery, electrons move as battery provides them energy to move.

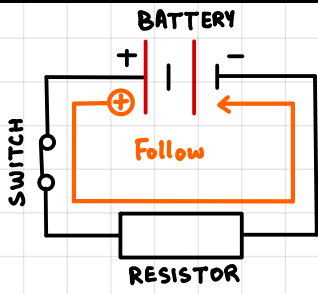
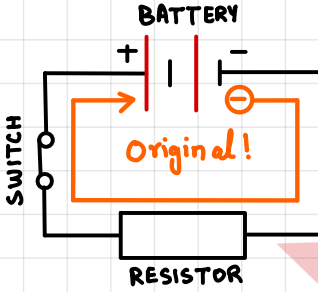


ALTERNATING Vs DIRECT CURRENT:

ALTERNATING CURRENT		- It doesn't have single direction of current. - It doesn't have constant size of current.
DIRECT CURRENT		- It has single direction of current. - It has constant size of current.



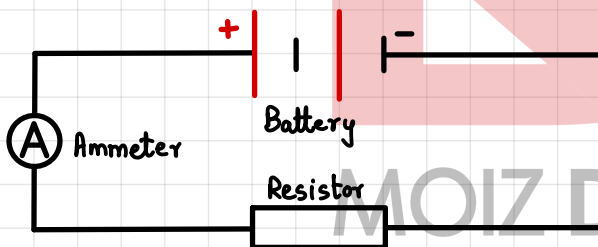
TYPES OF CURRENT:

CONVENTIONAL CURRENT		"Flow of positive charges from positive to negative terminal of the battery."
ELECTRON FLOW		"Flow of negative charges from negative to positive terminal of the battery."

MEASUREMENT OF CURRENT:

Circuit in which current has single path to flow is called series circuit.

Current can be measured by an ammeter (symbol $\textcircled{A}$). Ammeter is always placed in series.

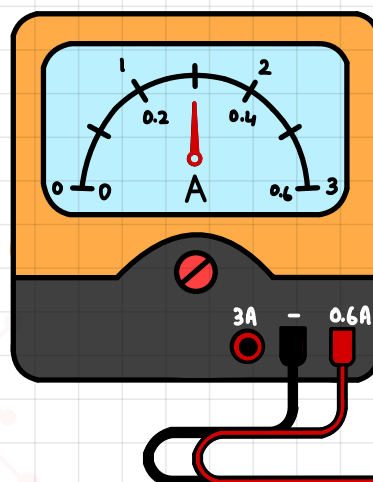


Ammeter can be placed anywhere in series as current remains same throughout the circuit.

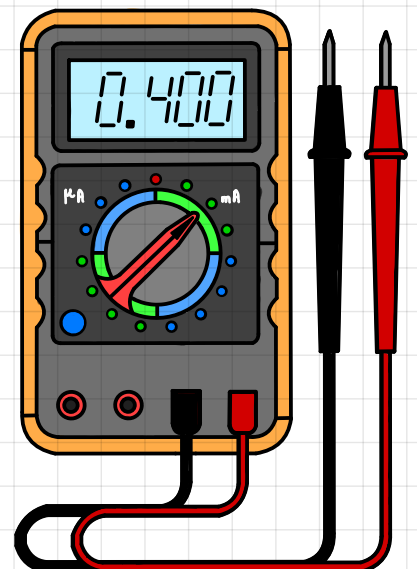
DIFFERENT RANGES OF AMMETER:

To measure current, always prefer to choose smaller range. Because, it is:

- More accurate.
- More sensitive.
- Easy to read.
- Digital meters are more accurate.



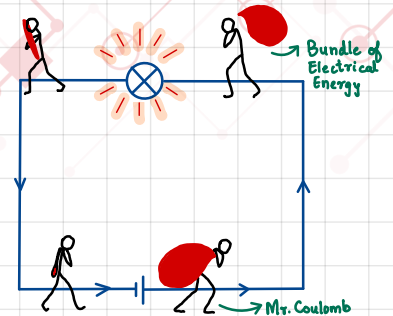
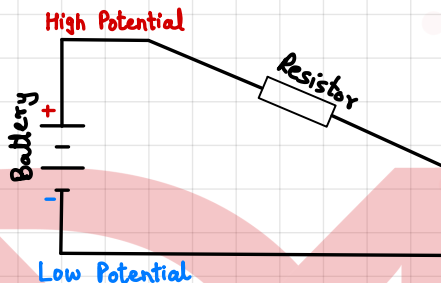
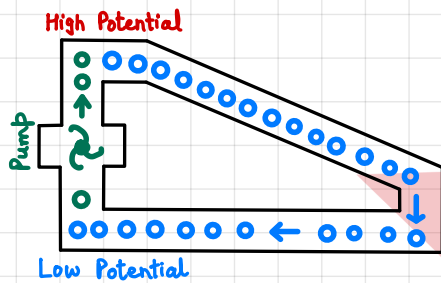
ANALOG



DIGITAL

EMF:

"Energy converted from non-electrical to electrical per unit charge which drives a charge throughout the circuit."

**FORMULA:**

$$\text{Emf} = \frac{\text{Energy}}{\text{Charge}} = \frac{\text{Work}}{\text{Charge}}$$

$$E = \frac{W}{q}$$

$$\text{Volt} = \frac{\text{Joule}}{\text{Coulomb}}$$

The S.I unit of emf is Volt (V).

POTENTIAL DIFFERENCE / VOLTAGE:

"Energy converted from electrical to non-electrical per unit charge which drives a charge through a single component."

FORMULA:

$$\text{Voltage} = \frac{\text{Work}}{\text{Charge}}$$

$$V = \frac{W}{q}$$

Its S.I unit is Volt (V).

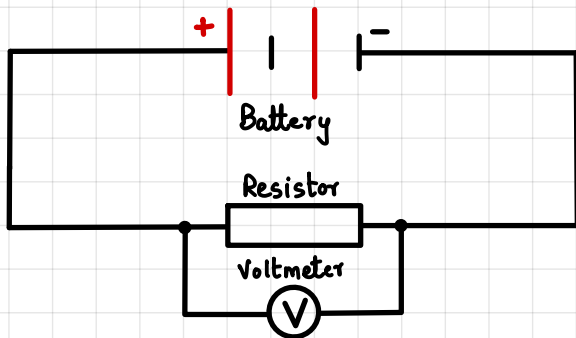
Both are measured by voltmeter placed in parallel.



MEASUREMENT OF EMF/P.D:

Circuit in which current has multiple paths to flow is called parallel circuit.

Emf & voltage both can be measured by voltmeter (Symbol V)
& voltmeter is always placed in parallel with the component.

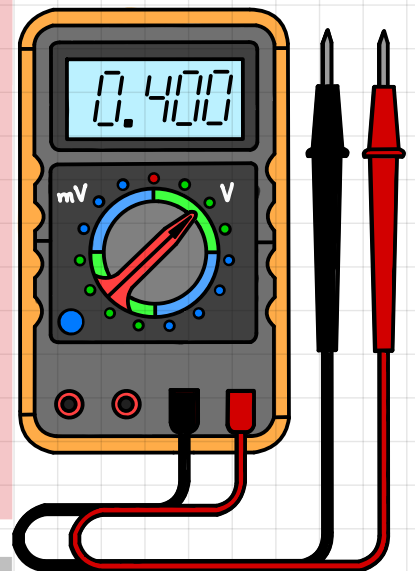
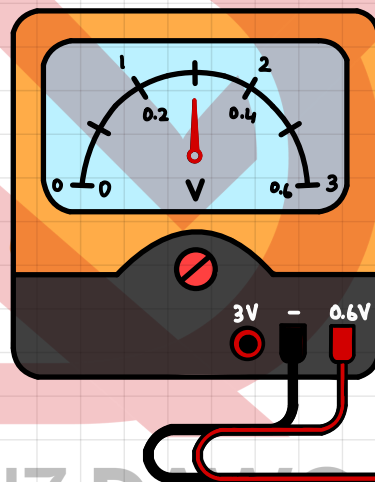


Voltmeter can be placed anywhere in parallel as voltage remains same in parallel.

DIFFERENT RANGES OF VOLTMETER.

To measure emf/P.d, always prefer to choose smaller range. Because, it is:

- More accurate.
- More sensitive.
- More readable.
- Digital meters are more accurate.



EMF SOURCES IN SERIES:

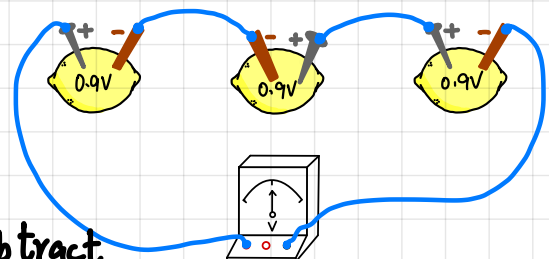
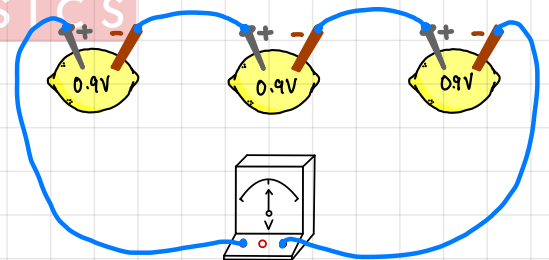
When emf sources are placed in series, they add up so the resultant emf is:

$$E = 0.9 + 0.9 + 0.9 = 2.7 \text{ V}$$

SPECIAL CASE:

When emf source are facing positive terminals together or vice versa, they subtract.

$$E = 0.9\cancel{\text{V}} - 0.9\cancel{\text{V}} + 0.9\text{V} = 0.9\text{V}$$

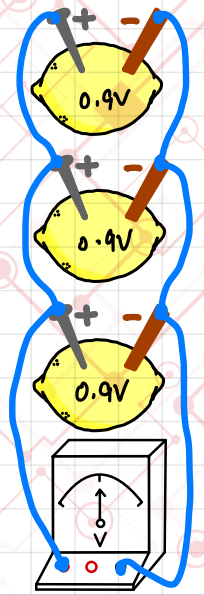


EMF SOURCES IN PARALLEL:

Since emf remains same in parallel so value of emf will be 0.9V.

The advantage of placing emf source in parallel are.

- Life of emf extends. or emf sources are less used up
- If one emf source runs down, other works properly.



4.2.4

Resistance

RESISTANCE:

"Opposition in the path of an electric current."

OR

* "Resistance is the ratio of voltage over current."

The S.I unit is Ohm (Ω omega).

FORMULA:

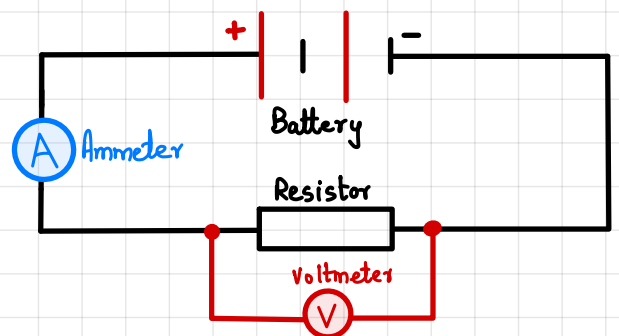
$$\text{Resistance} = \frac{\text{Voltage}}{\text{Current}}$$

$$R = \frac{V}{I}$$

$$\text{Ohm} = \frac{\text{Volt}}{\text{Ampere}}$$

MEASUREMENT OF RESISTANCE.

- Place an ammeter in series to measure current.
- Place a voltmeter in parallel to measure voltage across resistor.
- Using formula ' $R = \frac{V}{I}$ ', divide voltmeter reading by an ammeter to measure resistance.



FACTORS OF RESISTANCE:

- Resistance is directly proportional to the length ($R \propto L$)
- Resistance is inversely proportional to cross-sectional area ($R \propto \frac{1}{A}$).
- Change in Resistance is directly proportional to change in Temperature ($\Delta R \propto \Delta T$).

OHM'S LAW:

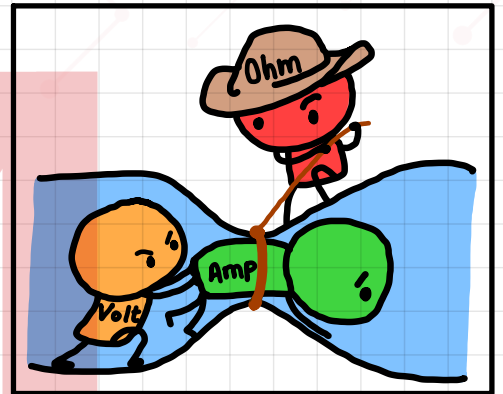
"Current is directly proportional to the potential difference provided that temperature remains constant."

$$I \propto V$$

$$I = kV$$

$$I = \frac{1}{R} V$$

$$R = \frac{V}{I}$$



OHMIC CONDUCTORS:

Conductors which follow Ohm's law are called Ohmic conductors. Their IV graph is a straight line.

E.g: metal conductor, Fixed Resistor etc.

NON-OHMIC CONDUCTORS:

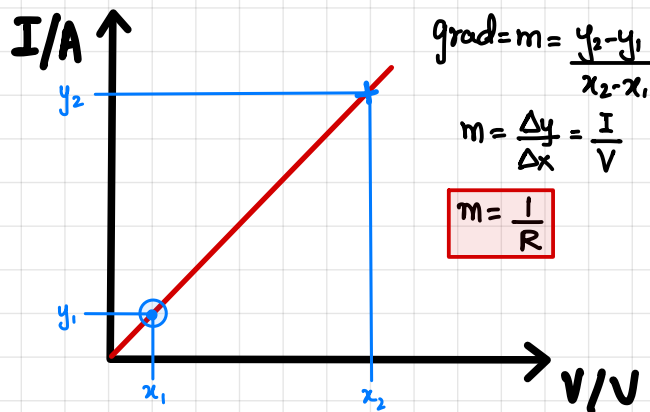
Conductors which don't follow Ohm's law are called non-Ohmic conductors. Their IV graph is not a straight line.

E.g: Filament Lamp, Thermistor, semiconductor diode etc.

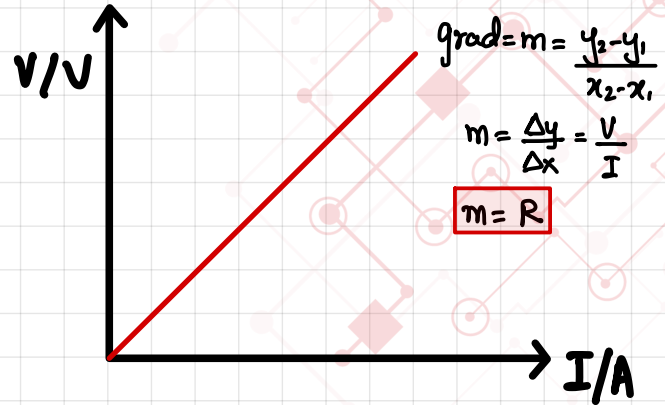
IV - GRAPH

VI - GRAPH

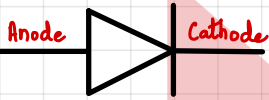
FIXED RESISTOR



The gradient of graph represents reciprocal of resistance.



The gradient of the graph represents resistance.

DIODE: 

Component which converts AC into DC is called a rectifier.

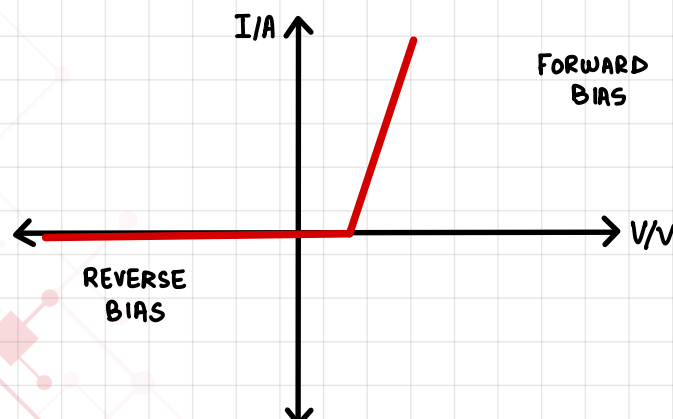
"A component which lets the current to pass through only in one direction." It works as a rectifier.

FORWARD BIAS: 

When battery is connected to diode such that negative is connected with cathode & positive is connected with anode. Diode works as a conductor.

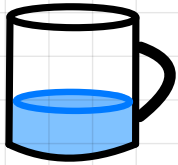
REVERSE BIAS: 

When battery is connected to diode such that negative is connected with anode & positive is connected with cathode. Diode works as an insulator.

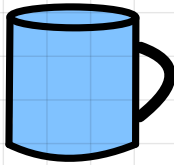


FILAMENT LAMP:

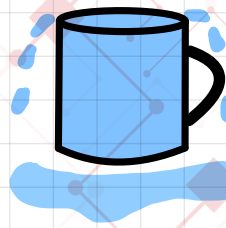
Resistance of filament increases when temperature increases.



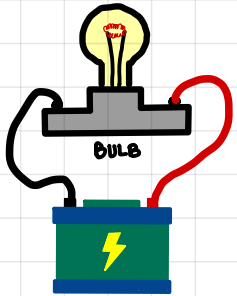
Capacity = 400ml
Water = 150ml
Water lost = 0 ml



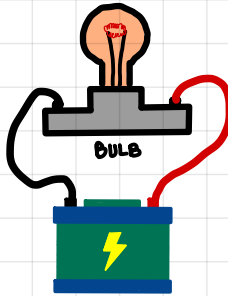
Capacity = 400ml
Water = 400ml
Water lost = 0 ml



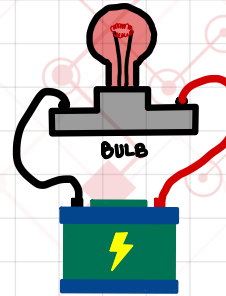
Capacity = 400ml
Water = 600ml
Water lost = 200ml



Rating = 6V
Battery = 3V
Brightness = Dim
Voltage lost = 0V



Rating = 6V
Battery = 6V
Brightness = Max
Voltage lost = 0V

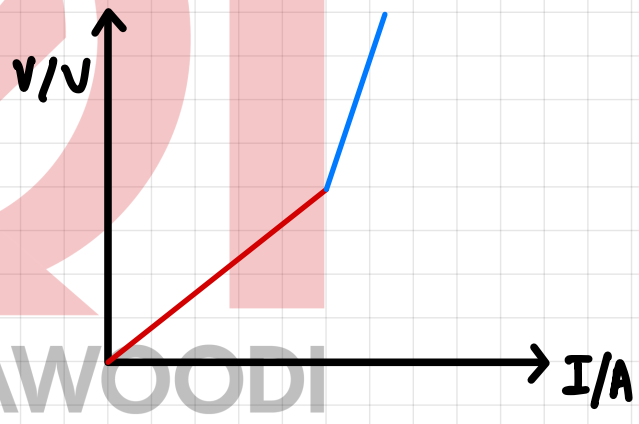
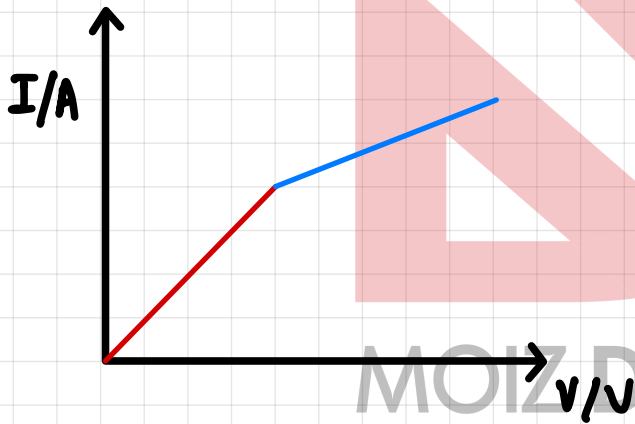


Rating = 6V
Battery = 8V
Brightness = Max
Voltage lost = 2V

IV - GRAPH

VI - GRAPH

FILAMENT LAMP

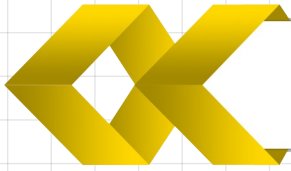


After maximum voltages, additional voltages are lost as heat.

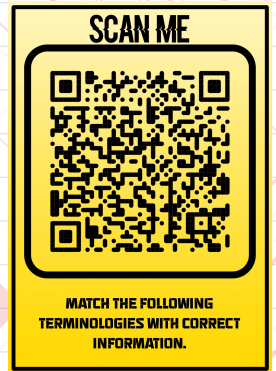
↑ Temperature

↑ Resistance

↓ Current



Match the following



Current is the	constant size & direction.
Volt is equal to	from non-electrical to electrical.
Direct current has	the opposition in the path of current.
Ammeter is always	flow of charges per unit time.
Emf changes energy	placed in series with component
Voltmeter is always	Joule per Coulomb.
Resistance is	inversely proportional to cross-sectional area.
Resistance increases	placed in parallel.
Resistance defines	when temperature is constant.
Ohm's law works	if length increases.

1 To charge an isolated metal sphere by induction, the following four processes are required.

- P The sphere is earthed by touching it.
- Q The earth connection is removed from the sphere.
- R A charged rod is brought close to the sphere.
- S The charged rod is removed.

In which order can these stages be carried out to charge the isolated metal sphere?

- A $P \rightarrow Q \rightarrow R \rightarrow S$
- B $P \rightarrow R \rightarrow S \rightarrow Q$
- C $R \rightarrow P \rightarrow Q \rightarrow S$
- D $R \rightarrow P \rightarrow S \rightarrow Q$

[MJ2011/P11/Q27]

2 The potential difference (p.d.) across a 10Ω resistor is $5V$.

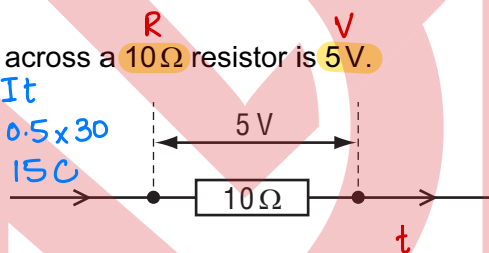
$$R = \frac{V}{I}$$

$$I = \frac{V}{R} = \frac{5}{10} = 0.5A$$

$$q = It$$

$$= 0.5 \times 30$$

$$q = 15C$$



How much charge passes through the 10Ω resistor in 30 seconds?

- A 2C
- B 15C
- C 60C
- D 1500C

[MJ2011/P11/Q28]

3 Which changes both cause a decrease in the resistance of a copper wire?

	size of wire	temperature of wire
A	decrease in length ✓	lower ✓
B	increase in length ✗	lower ✓
C	decrease in thickness ✗	higher
D	increase in thickness ✓	higher

$$R \propto L$$

$$R \propto \frac{1}{A}$$

$$\Delta R \propto \Delta T$$

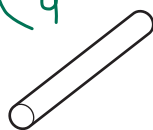
[MJ2011/P11/Q29]

- 4 The resistance of a cylindrical wire P is 80Ω . A second wire Q is made from the same material. The cross-sectional area of Q is four times that of P. The length of Q is twice the length of P.

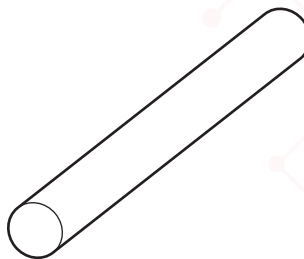
$$R \propto \frac{1}{A}$$

If A is 4 times
So R is $\frac{1}{4}$

$$\left(\frac{80}{4} = 20\Omega\right)$$



wire P



wire Q

$$R \propto L$$

If "L" is 2 times
"R" is double
 $20 \times 2 = 40\Omega$

What is the resistance of Q?

- A 10Ω **B 40Ω** C 160Ω D 640Ω [ON2011/P11/Q28]

Friction

- 5 A polythene rod is rubbed with a duster. The duster then attracts small pieces of paper.

Are the rod and the duster charged or uncharged?

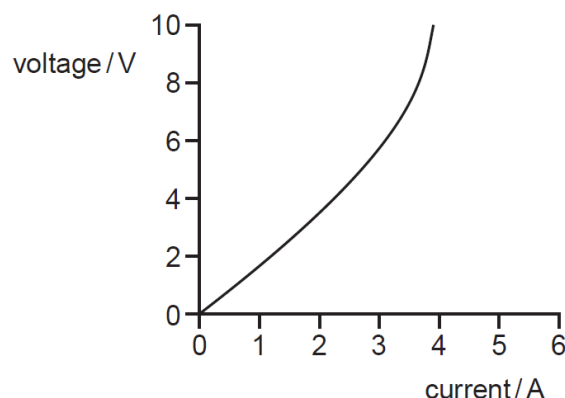
	rod	duster
A	charged ✓	charged ✓
B	charged ✓	not charged
C	not charged	charged ✓
D	not charged	not charged

[MJ2012/P11/Q27]

- 6 The voltage/current graph for a filament lamp is shown.

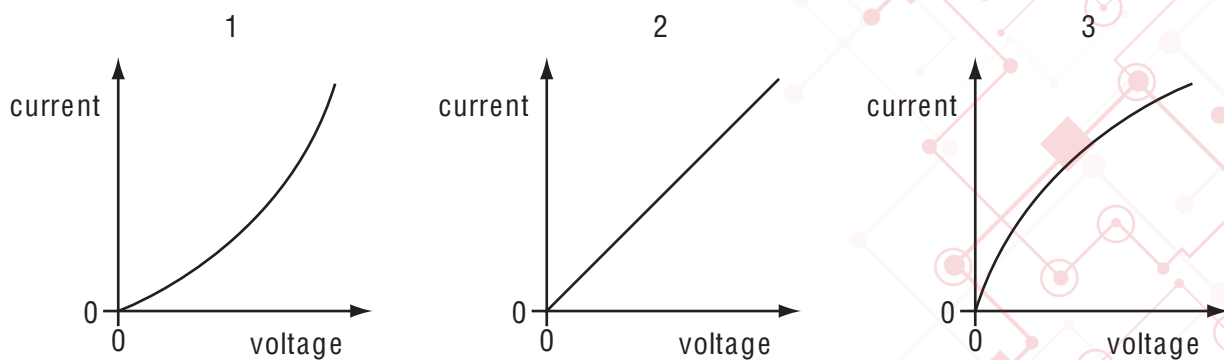
The voltage across the lamp increases. What happens to the temperature of the lamp and the resistance of the lamp?

	temperature	resistance
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases



[MJ2012/P11/Q28]

7 The current/voltage graphs are for different electrical components.

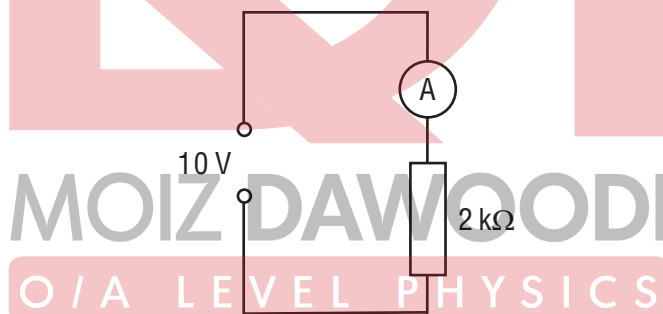


Which graph is for a resistor at constant temperature and which is for a filament lamp?

	resistor	lamp
A	1	2
B	2	1
C	2	3
D	3	2

[MJ2012/P12/Q28]

8 The diagram shows an ammeter connected in a circuit.



What is the current in the ammeter?

A 5 mA

B 20 mA

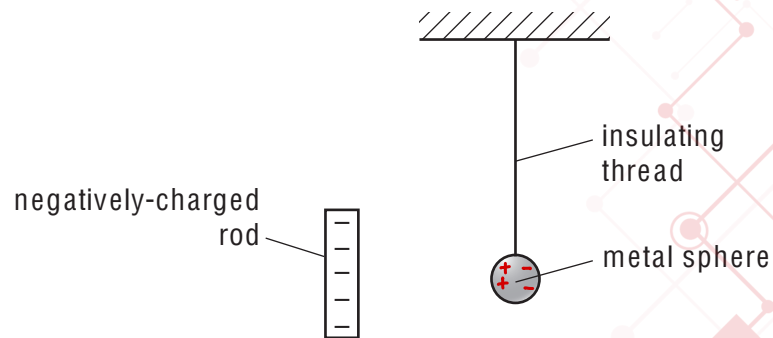
C 0.2 A

D 5 A

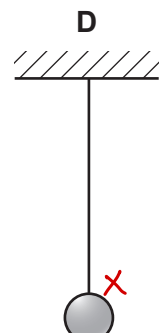
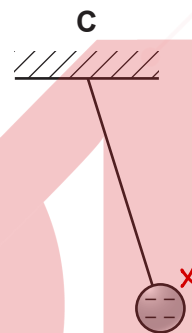
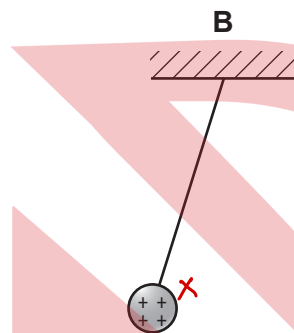
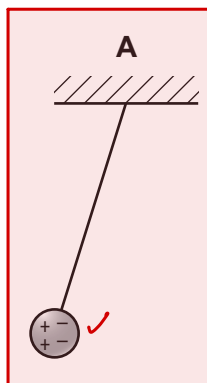
[MJ2012/P12/Q29]

- 9 A small **uncharged metal** sphere hangs from an insulating thread.

A negatively-charged rod moves close to the sphere.



Which diagram shows the charges on the sphere and its final position?



[ON2012/P12/Q27]

- 10 A metal wire has length l and cross-sectional area A .

What is the resistance proportional to?

A $A + l$

B $\frac{A}{l}$

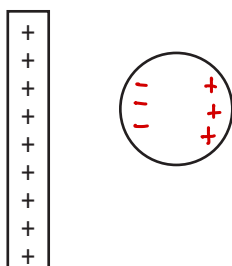
C $\frac{l}{A}$

D $l \times A$

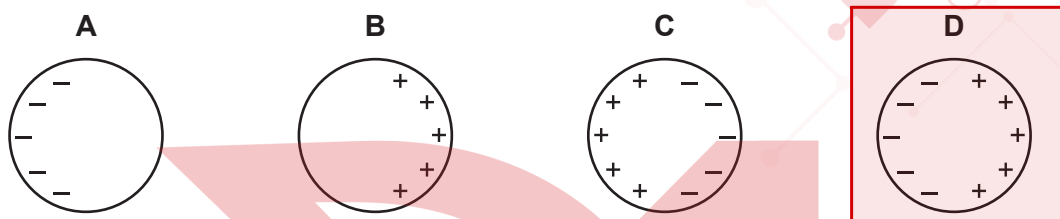
MOIZ DAWOODI
O / A LEVEL PHYSICS

[ON2012/P12/Q28]

- 11 A positively-charged rod is held close to, but not touching, an isolated metal sphere.



Which diagram shows the charges that are induced on the sphere?



[MJ2013/P11/Q28]

- 12 A lightning flash carries 40 C of charge and lasts for 5.0 ms.

What is the average current in the flash?

- A 0.20 A B 8.0 A C 200 A

D 8000 A

$$I = \frac{q}{t} = \frac{40}{5 \times 10^{-3}} = 8000 \text{ A}$$

[MJ2013/P11/Q29]

- 13 Which of the following is equivalent to 1 V?

- A 1 J/A
B 1 J/C
C 1 J/s
D 1 J/W

[MJ2013/P11/Q30]

- 14 A student rubs a rod held in his hand.

charging due to friction = Insulator

Which action causes the rod to gain a large electrostatic charge?

- A rubbing an iron rod with a steel magnet
B rubbing an iron rod with a woollen duster
C rubbing a polythene rod with a steel magnet
D rubbing a polythene rod with a woollen duster

[MJ2013/P12/Q31]

- 15 A lightning flash carries 40 C of charge and lasts for 5.0 ms.

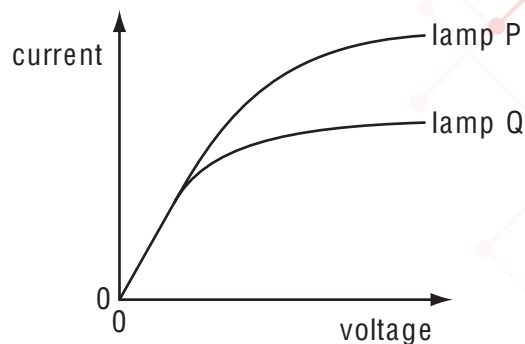
Repeated

What is the average current in the flash?

- A 0.20 A B 8.0 A C 200 A D 8000 A

[MJ2013/P12/Q32]

16 The diagram shows the current-voltage graph of two filament lamps.



Which statement is correct?

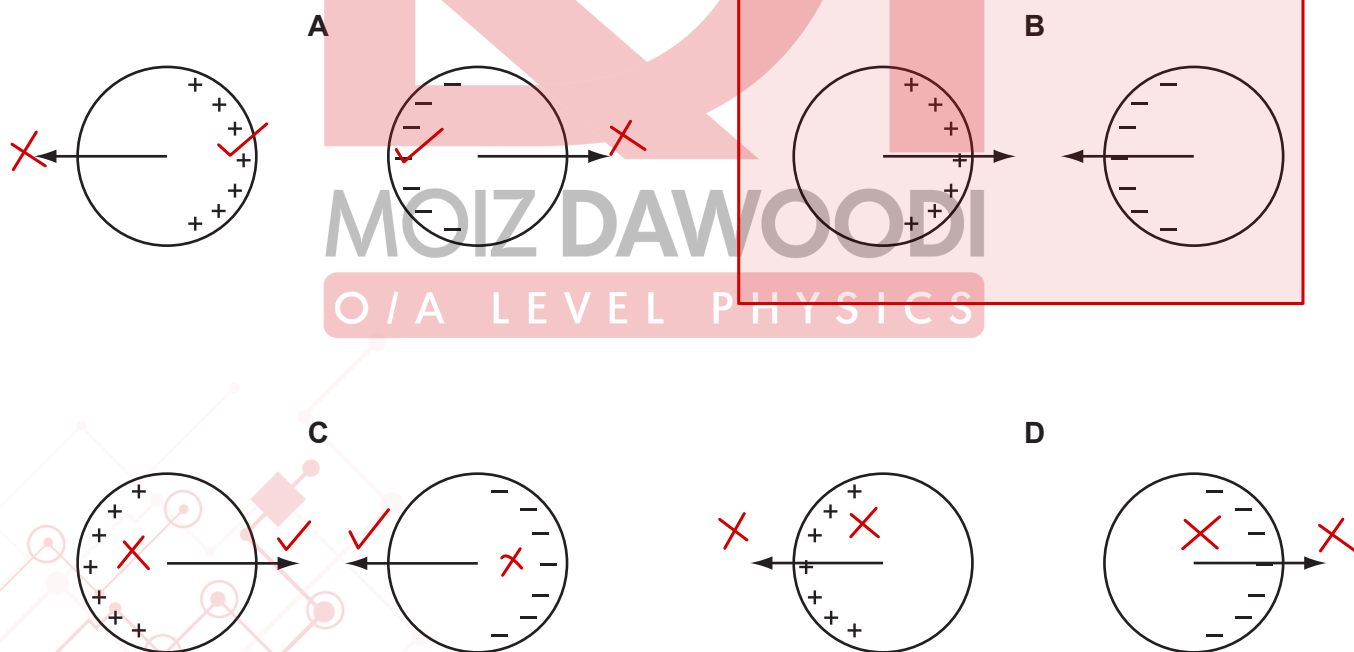
- A** Lamp P has a lower resistance than lamp Q at all currents.
- B** Lamp P has the same resistance as lamp Q at low currents.
- C** Lamp P has the same resistance as lamp Q at high currents.
- D** Lamp P has a higher resistance than lamp Q at all currents.

[MJ2013/P12/Q33]

17 Two charged conducting spheres are placed close to one another.

One sphere is positively charged and the other is negatively charged.

Which diagram shows the distribution of charges and the forces acting on the spheres?



[ON2013/P11/Q26]

18 Which row shows a conductor and an insulator?

	conductor	insulator
A	aluminium	copper
B	aluminium	glass
C	plastic	copper
D	plastic	glass

[ON2013/P11/Q27]

19 Which of the following is equivalent to one coulomb?

A one ampere second

$$q = It$$

B one ampere per volt

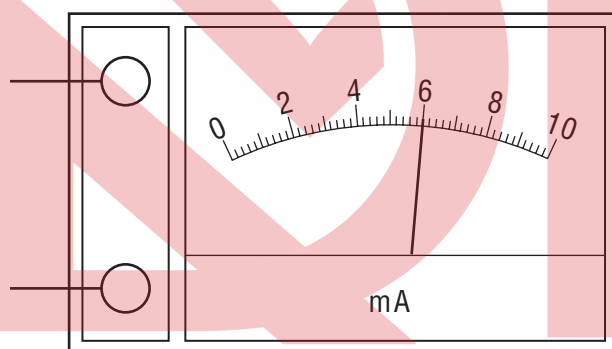
$$C = As$$

C one volt ampere

D one volt per ampere

[ON2013/P11/Q28]

20 An ammeter is used to measure the current in a 300Ω resistor. The ammeter is shown below.



What is the potential difference across the resistor?

A 0.050 V

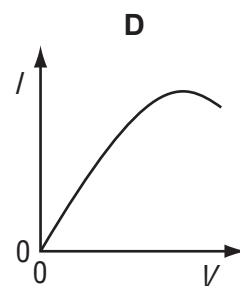
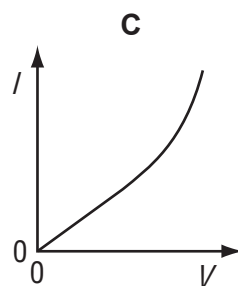
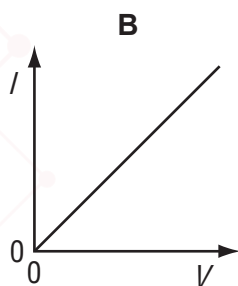
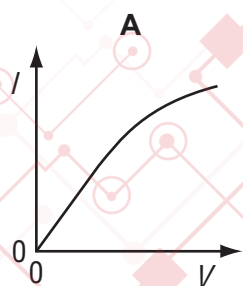
B 1.8 V

C 50 V

D 1800 V

[ON2013/P12/Q28]

21 Which is the current/voltage (I/V) graph of a filament lamp?



[ON2013/P12/Q30]

$$R \propto \frac{1}{A}$$

$$2A \rightarrow \frac{1}{2}R \rightarrow \frac{64}{2} = 32 \Omega$$

$$R \propto L$$

$$4L \rightarrow 4R \rightarrow 4 \times 16 \rightarrow 64 \Omega$$

- 22 A piece of wire has a resistance of 16Ω .

Another wire made from the same metal has four times the length and twice the cross-sectional area.

What is the resistance of the wire?

- A 8Ω **B 32Ω** C 96Ω D 128Ω [MJ2014/P11/Q30]

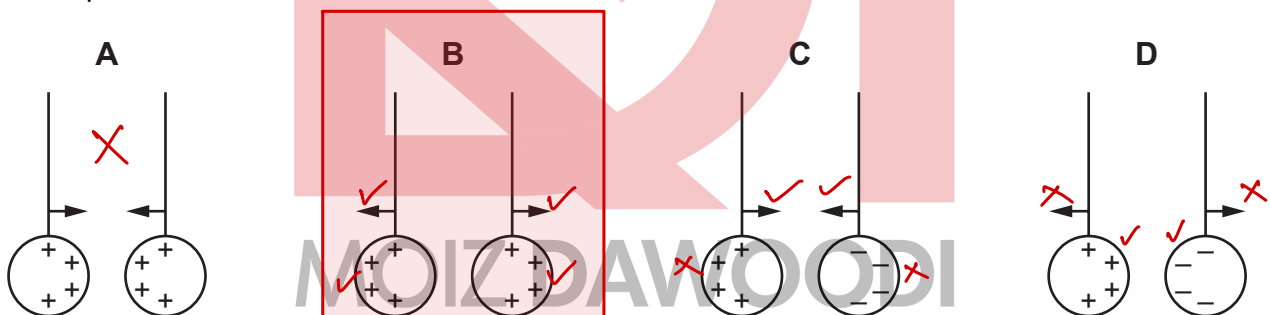
- 23 Which row shows an electrical conductor and an insulator?

	electrical conductor	insulator
A	aluminium	rubber
B	copper	aluminium
C	plastic	copper
D	rubber	plastic

[MJ2014/P12/Q31]

- 24 Two charged metal spheres are suspended close to each other.

Which diagram shows the charge distribution on the spheres and the directions of the forces on the spheres?



[ON2014/P12/Q23]

- 25 Which list contains only electrical insulators?

- A glass, plastic, rubber
B glass, plastic, steel
C glass, rubber, steel
D plastic, rubber, steel

$$I = \frac{q}{t} = \frac{7.5}{5} = 1.5 \text{ A}$$

[ON2014/P12/Q24]

- 26 A charge of 7.5 C flows through a resistor in 5.0 s . A student has ammeters with different ranges that he can use to measure the current in the resistor.

Which ammeter range is the most appropriate?

- ~~A~~ $0-1 \text{ A}$ **B $0-2 \text{ A}$** C $0-5 \text{ A}$ ~~D~~ $0-40 \text{ A}$ [MJ2015/P11/Q26]

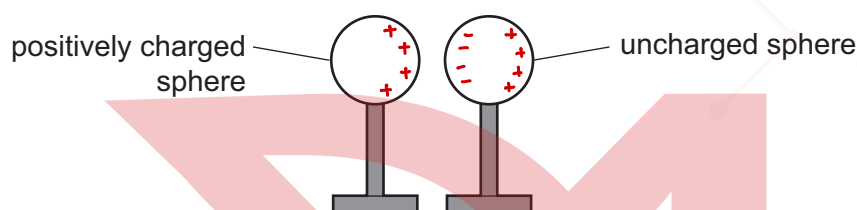
- 27 A polythene rod becomes charged when rubbed with a dry woollen cloth.

The charge caused is a result of

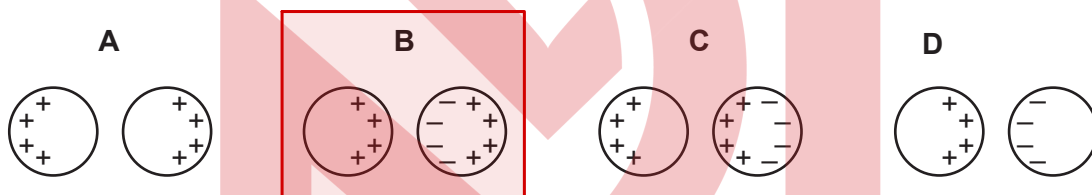
- A friction producing a movement of electrons.
- B friction producing a movement of protons.
- C magnetism producing a movement of electrons.
- D magnetism producing a movement of protons.

[ON2015/P11/Q24]

- 28 A positively charged insulated metal sphere is brought close to an uncharged insulated metal sphere.



Which diagram shows the charge distribution on the spheres?



[ON2015/P11/Q25]

- 29 The flash on a camera operates when a capacitor discharges through a flash-tube. In 3.2 ms, a charge of 6.0 C passes through the flash-tube.

What is the average current in the flash-tube?

- A 0.019 A
- B 1.9 A
- C 19 A
- D 1900 A

$$I = \frac{q}{t} = \frac{6}{3.2 \times 10^{-3}} = 1875 \text{ A}$$

[ON2015/P11/Q26]

- 30 Which physical quantity is produced by a calculation where a charge is multiplied by a potential difference (p.d.)?

- A current
- B electromotive force (e.m.f.)
- C energy
- D power

$$? = q \times V$$

$$V = \frac{W}{q}$$

$$\text{Energy} = W = qV$$

[ON2015/P11/Q27]

31 Which row correctly names an electrical conductor and an insulator?

	electrical conductor	insulator
A	aluminium	iron
B	iron	rubber
C	plastic	aluminium
D	rubber	plastic

[ON2015/P12/Q29]

P 1

Mark Scheme

1	B	16	B
2	A	17	B
3	C	18	B
4	B	19	A
5	D	20	B
6	C	21	A
7	C	22	B
8	A	23	A
9	A	24	B
10	C	25	A
11	D	26	C
12	D	27	A
13	A	28	B
14	D	29	D
15	D	30	C

- 1 Fig. 1.1 shows a nylon rod being rubbed with a woollen cloth.



Fig. 1.1

The rod becomes negatively charged.

- (a) (i) Explain how rubbing the nylon rod makes it negatively charged.

Due to friction, electrons of wool jumped towards rod.
As rod has gained electrons, it becomes negatively charged. [1]

- (ii) State what happens to the cloth as the rod becomes negatively charged.

Positively charged.

[1]

- (b) Fig. 1.2 shows the negatively-charged rod placed near to a conducting sphere. The sphere is suspended by an insulating thread.

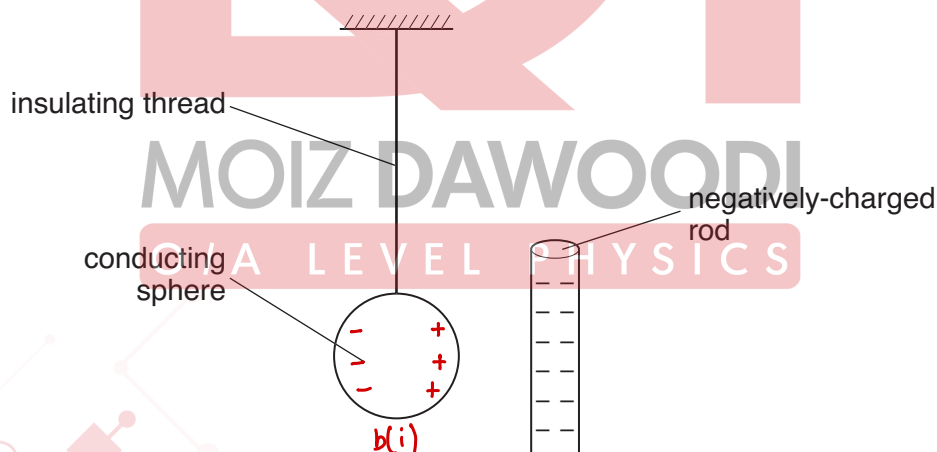


Fig. 1.2

- (i) On Fig. 1.2, draw the distribution of positive and negative charges on the sphere. [2]
- (ii) Explain why the sphere is attracted to the rod.

Since distance b/w unlike charges is less so force of attraction is greater than repulsion.

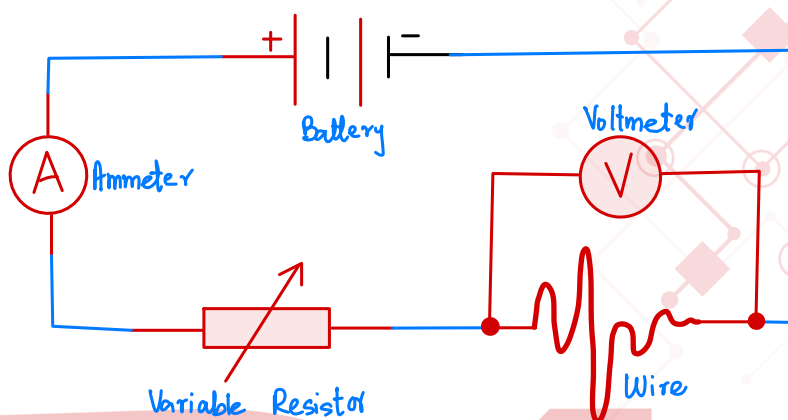
[2]

[ON2011/P21/Q5]

- 2 An electrical engineer ^{voltmeter Parallel} measures the potential difference across a length of metal wire and the ^{Variable Resistor} current in the wire. He does this for different values of the current.

^{Ammeter Series}

- (a) (i) Draw a labelled circuit diagram of a circuit that enables the engineer to do this.



[3]

- (ii) Describe how the circuit is used.

Ammeter measures current. Voltmeter measures potential difference of metal wire. using variable resistor, value of current can be altered. Different length of wires are used to see comparison.

[2]

- (iii) State the equation that defines resistance.

$$R = V/I$$

[1]

- (b) The engineer performs the experiment and notices that the potential difference across the wire is directly proportional to the current in it. He calculates the resistance of the wire and plots a graph of the resistance against the current.

On Fig. 2.1, sketch the shape of this graph.

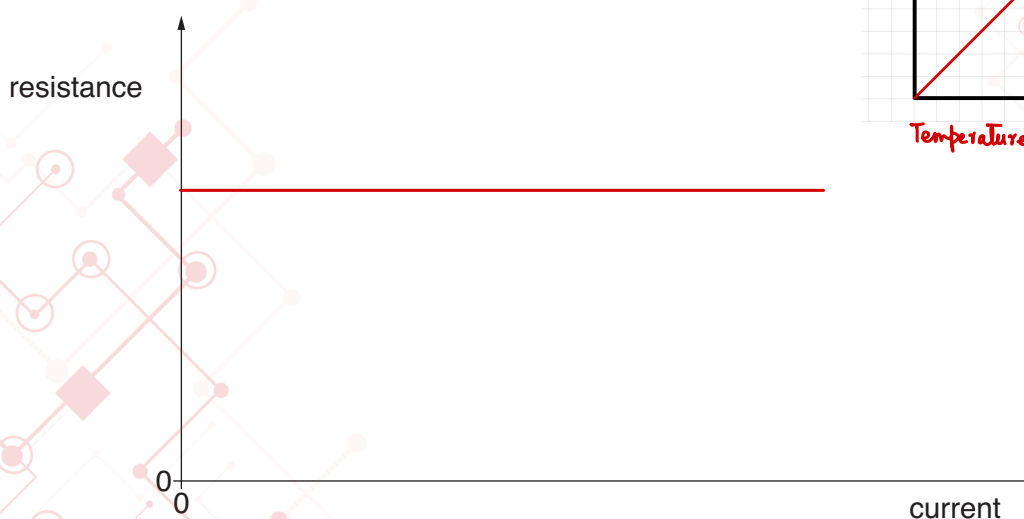
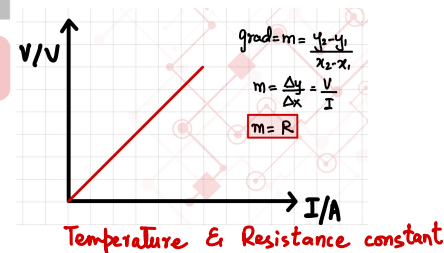


Fig. 2.1

[1]



[ON2011/P22/Q6]

3 Fig. 3.1 shows two charged metal spheres.

One sphere has a positive charge and the other sphere has a negative charge.

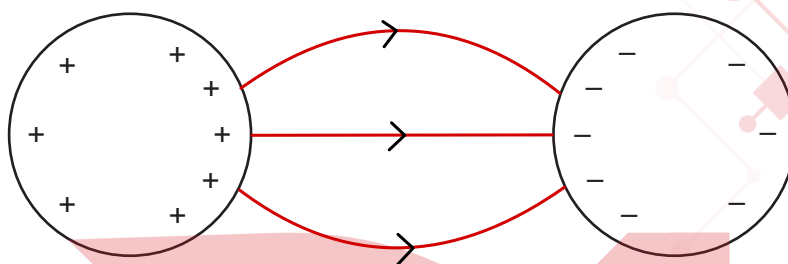


Fig. 3.1

(a) On Fig. 3.1, draw the electric field between the two spheres. [2]

(b) The negative charge on the sphere is removed and a wire is used to connect the positive sphere to earth. The charge on the positive sphere decreases from $4.8 \times 10^{-9} \text{ C}$ to zero in a time of $2.0 \times 10^{-6} \text{ s}$.

Calculate the average current in the wire.

$$I = \frac{q}{t} = \frac{4.8 \times 10^{-9}}{2 \times 10^{-6}}$$

O/A LEVEL PHYSICS

current = $2.4 \times 10^{-3} \text{ A}$ [2]

[MJ2012/P22/Q7]

Question 1
Static Electricity

On a stormy day, a large, positively-charged cloud is above a tree.
An electrical charge is induced on the tree as charged particles flow through it.

What is the charge induced on the tree and how do the charged particles move?

	charge on tree	movement of charged particles through tree
A	negative ✓	negatively charged particles move down the tree
B	negative ✓	negatively charged particles move up the tree ✓
C	positive	positively charged particles move down the tree
D	positive	positively charged particles move up the tree

- 4 Fig. 4.1 is a graph of current against potential difference (p.d.) for a length of metal wire.

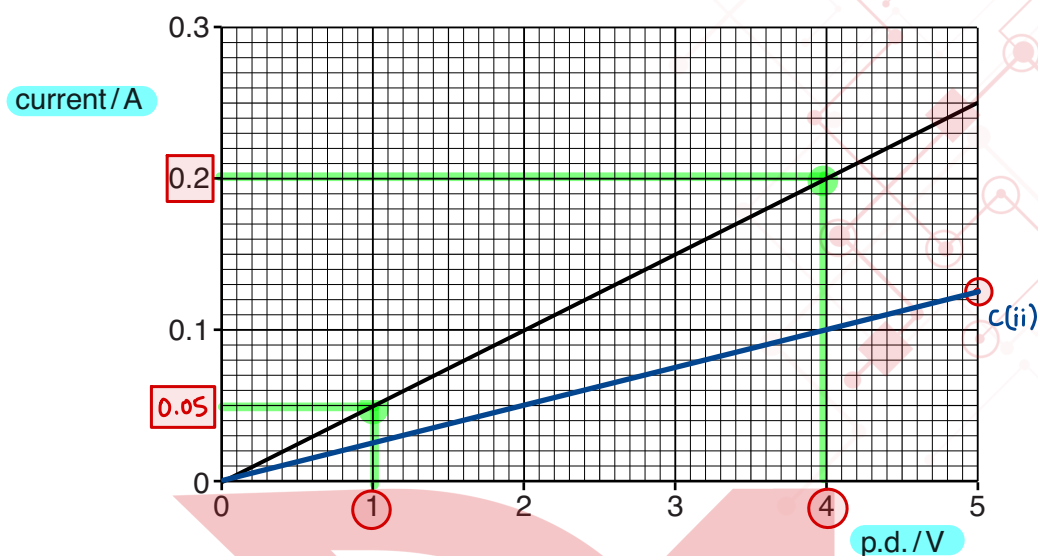


Fig. 4.1

- (a) The metal wire obeys Ohm's law. State Ohm's law in words.

Current is directly proportional to the voltage provided that temperature remains constant.

[2]

- (b) Calculate the resistance of the metal wire.

$$\frac{1}{R} = \text{grad} = \frac{\Delta I}{\Delta V} = \frac{0.2 - 0.05}{4 - 1} = 0.05$$

$$R = \frac{V}{I} = \frac{4}{0.2}$$

$$\frac{1}{R} = 0.05$$

$$R = \frac{1}{0.05} = 20 \Omega$$

OR

$$\text{resistance} = 20 \Omega$$

[2]

- (c) A new wire is made from the same metal as the original wire. The two wires have the same length. The cross-sectional area of the new wire is half that of the original wire.

- (i) Calculate the resistance of the new wire.

$R \propto \frac{1}{A} \rightarrow$ Area half
Resistance double

$$20 \times 2$$

$$\text{resistance} = 40 \Omega$$

[1]

- (ii) On Fig. 4.1, draw a line to show how current varies with p.d. for the new wire.

[2]

[MJ2013/P21/Q6]

- 5 A student rubs a polythene rod with a dry cloth. The polythene rod becomes negatively charged and the cloth becomes positively charged.

(a) Describe, in terms of the movement of electrons, what happens during the charging process.

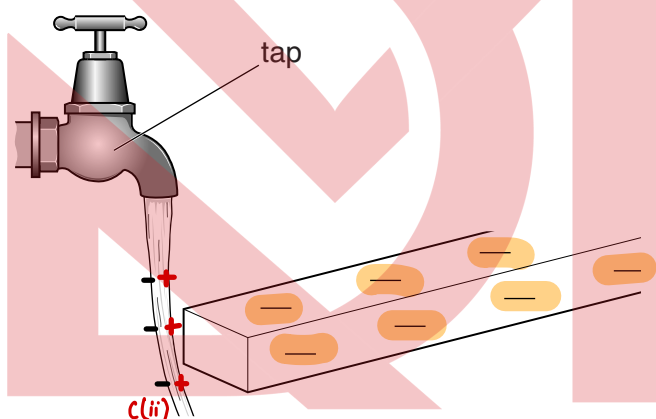
Electrons of cloth jumped towards rod. As rod has gained electrons, it becomes negative & cloth has lost electrons, it becomes positive. [2]

(b) There is an electric field around the charged rod.

Explain what is meant by an electric field.

The region around a charge in which charge experiences electric force. [1]

(c) The charged rod is placed near a stream of water from a tap, as shown in Fig. 5.1.



MOIZ DAWOODI
Fig. 5.1

The stream of water bends because the rod is charged.

(i) Explain how the charge on the rod causes positive and negative particles in the water to move.

Rod attracts positive towards it as unlike charges attract & repels negative charges as like charges repel. [2]

(ii) On Fig. 5.1, draw the charge present on the stream of water. [1]

[MJ2013/P22/Q5]

- 6 A positively charged sphere C is brought close to a small, uncharged metal sphere U. Sphere U is suspended from an insulating thread, as shown in Fig. 6.1.

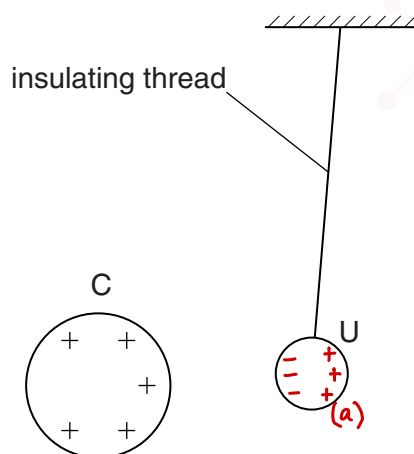


Fig. 6.1

- (a) On Fig. 6.1, draw the induced charges on sphere U. [1]
- (b) Sphere C is moved towards sphere U until the spheres touch. Sphere U is then repelled by sphere C, as shown in Fig. 6.2. The charges on C and U are not shown.

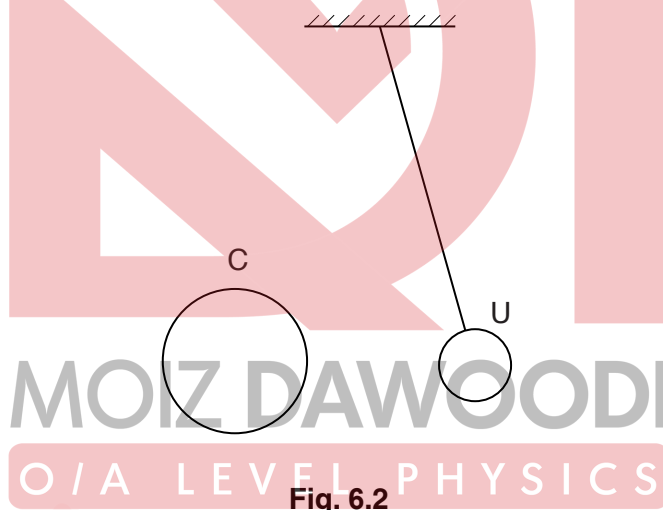
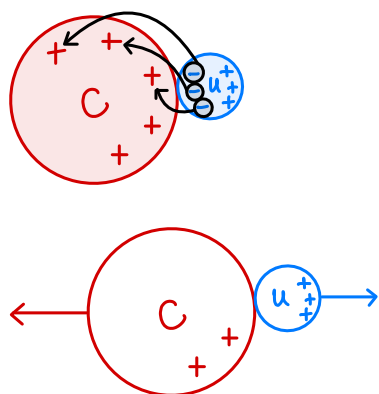


Fig. 6.2

- (i) State and explain what happens to the charge on the two spheres as they touch.

charge on C: Positive
But less than before [2]

charge on U: Positive [1]
[3]

- (ii) Explain why U is repelled by C.

Both are like charged which repel each other. [1]

[MJ2014/P21/Q5]

- 7 The base of a storm cloud is negatively charged. Fig. 7.1 shows the cloud above flat ground.

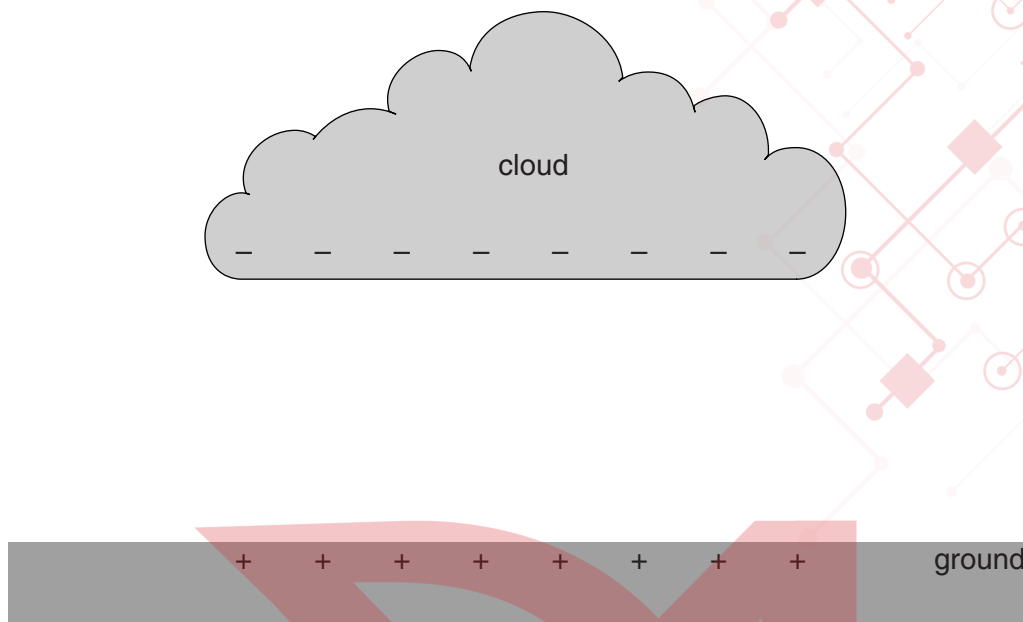


Fig. 7.1

- (a) The cloud causes the ground beneath it to become positively charged.

Explain, in terms of the particles involved, how the ground becomes positively charged.

Electrons of ground are repelled by electrons in the cloud to the depth. Leaving the ground positively charged.

[2]

- (b) In the space between the negative charge on the cloud and the positive charge on the ground, there is an electric field.

State what is meant by an electric field.

The region around a charge in which charge can experience electric force.

[1]

- (c) A lightning strike takes place. In 0.0015s, a charge of 180C passes between the cloud and the ground.

Calculate the average current in the lightning strike.

$$I = \frac{q}{t} = \frac{180}{0.0015}$$

current = 120000 A [2]

[ON2014/P21/Q6]

- 8 When a balloon is rubbed on hair, the balloon becomes negatively charged. The balloon is shown in Fig. 8.1.

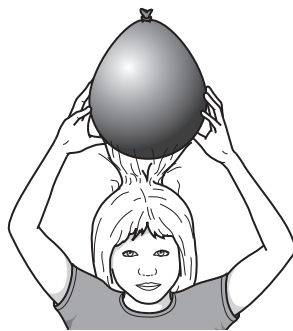


Fig. 8.1

- (a) Explain how rubbing causes the balloon to become negatively charged.

Electrons of hair jump towards balloon, as balloon has gained electrons, it becomes negatively charged.

[2]

- (b) Explain why the hair is pulled towards the balloon.

As hair has lost electrons, it becomes positively charged. Hair & balloon are unlike charged so unlike charges attract.

[2]

- (c) Explain why it is important that the balloon is made from an electrical insulator.

Insulator can't move their energetic electrons within themselves so they utilize energy by transfer of electrons toward other body.

[1]

[MJ2015/P22/Q5]

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – October/November 2011	5054	21

- 5 (a) (i) **electrons** move to the rod B1
- (ii) becomes positively-charged/loses electrons B1
- (b) (i) positives on right **and** negatives on left M1
equal numbers(at least 2) **and** roughly symmetrical A1
- (ii) positive charges attracted B1
attraction larger than repulsion **or** positives closer (than negatives to rod) B1 [6]

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
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- 6 (a) (i) power supply, (wire/resistor/bulb) and ammeter in series B1
voltmeter across wire/resistor/bulb **labelled/clear** B1
variable power supply **or** rheostat in series **or** potentiometer B1
correct symbols **or** labelled throughout
- (ii) read ammeter **and** voltmeter / measure voltage **and** current B1
vary power supply/rheostat/current B1
- (iii) ($R =$) V/I (ign. V/A) B1
- (b) horizontal line **and** above axis B1 [7]

Page 4	Mark Scheme: Teachers' version	Syllabus	Paper
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- 7 (a) three lines from one sphere to the other **and** some lines should spread out as B1
they leave one sphere **and** come together nearing the other C S
correct direction on at least one line and none wrong B1
- (b) ($I =$) Q/t in any form numerical or algebraic C1
 $2.4 \times 10^{-3} \text{ A}$ A1 [4]

Page 3	Mark Scheme	Syllabus	Paper
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- 6 (a) current is directly proportional to voltage (**accept** voltage/current = constant, but **not** just = R) B1
if temperature/physical conditions constant B1
- (b) ($R=$) V/I in any form algebraic or using any value of V and I from graph C1
 20Ω A1
- (c) (i) 40Ω or $2 \times$ (b) B1
- (ii) straight line graph through origin below given line ecf (b) (e.g. **accept** above line if $R < 20$) M1
goes through 0.1 A at 4 V ecf (b) (e.g. allow through 0.2 A at 2 V if $R = 10\Omega$) A1 [7]

Page 3	Mark Scheme	Syllabus	Paper
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- 5 (a) electrons move onto polythene / rod B1
electrons/negative charge move off cloth B1
- (b) (region of space) where force is exerted on a charge B1
- (c) (i) unlike charges attract B1
or (rod) attracts +ve charge/ions/particles
repels like charge B1
or (rod) repels –ve charge/ions/electrons/particles
- (ii) (net) positive charge on water near rod B1 [6]

Page 3	Mark Scheme	Syllabus	Paper
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- 5 (a) **equal numbers** (5 or less) of negative charges on left **and** positive charges on right [B1]
- (b) (i) C becomes less positive/less charged [B1]
U becomes (completely) positive [B1]
electrons/negative charge flows from U to C or + (on C) and – (on U)
cancel/neutralise [B1]
- (ii) like charges repel or both have same charge or both positive [B1]

Page 3	Mark Scheme	Syllabus	Paper
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- 6 (a) **electrons** repelled by cloud (leaving ground positive) **not** positive charge / protons move
like charges repel **or** electrons negative

B1
B1

- (b) (region) where (electric) charge experiences a force

B1

- (c) ($I =$) Q/t **or** $180/0.0015$
 1.2×10^5

C1

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2015	5054	22

- 5 (a) negative charge moves from hair/person/head to balloon
electrons move from hair/person/head to the balloon

C1
A1

- (b) hair is positive (at end)
opposite charges attract
or positive and negative attract

B1
B1

- (c) **charges / electrons**
- don't flow away
 - aren't conducted (to earth/person)
stay on balloon/on insulator

B1

MOIZ DAWOODI
O / A LEVEL PHYSICS